

Algorithmic Fashion: From Raw Pixels to Personalized Recommendations

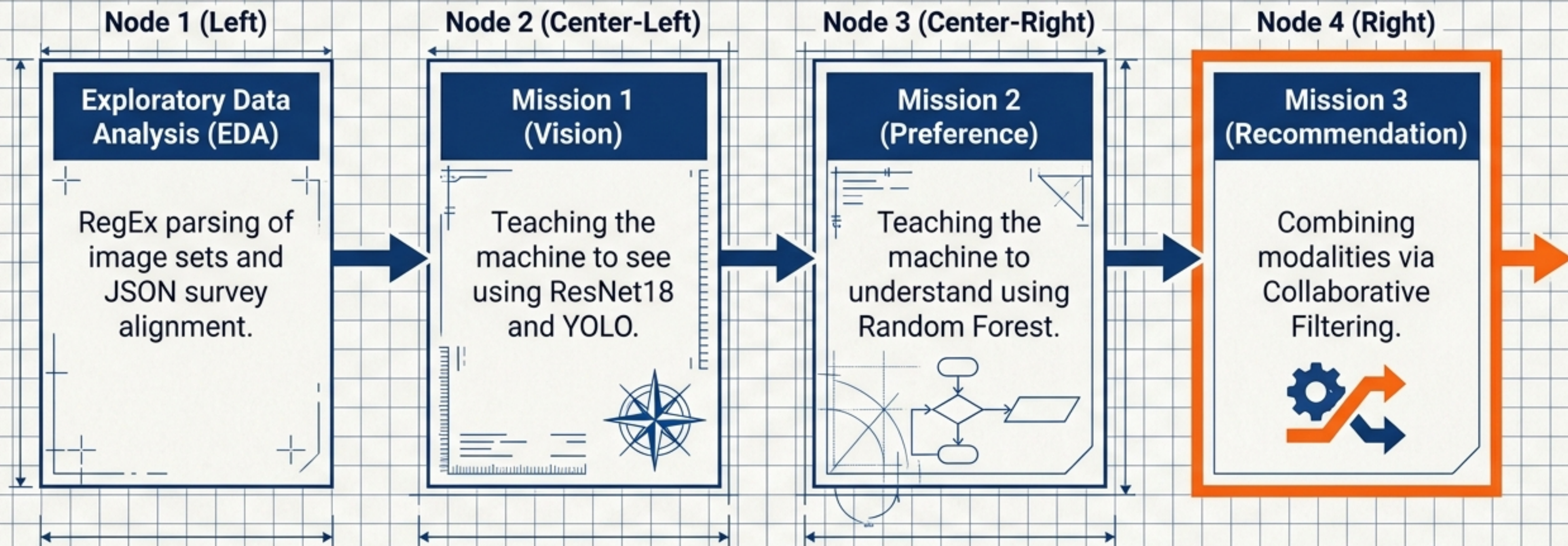
An End-to-End Multimodal Recommendation
Pipeline for the Data Creator Camp (DCC)

Data Architecture | Computer Vision |
Collaborative Filtering

System Efficacy: 92.11%
Classification Accuracy



The Data-to-Value Pipeline



Establishing the Data Foundation

Automated extraction and structuring of visual attributes to construct a balanced training environment.

Input: Raw Image Files

e.g., W_04972_90_kitsch_W.jpg



Process: RegEx Parsing

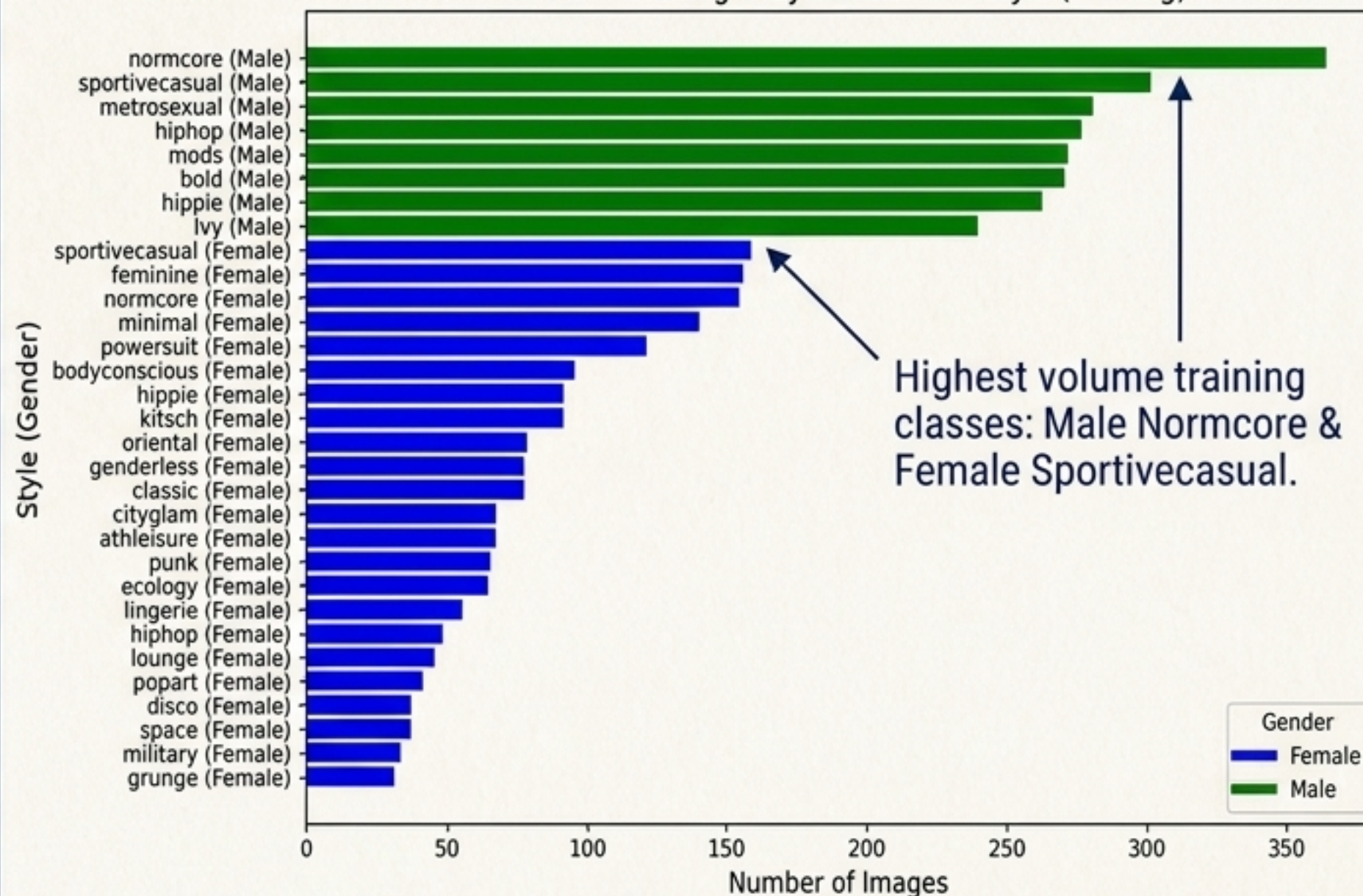
"W" or "T" _ [digits] _ [letters] _ [letters] .jpg



Output: Structured Data Frame

Gender, Style, Count = 1

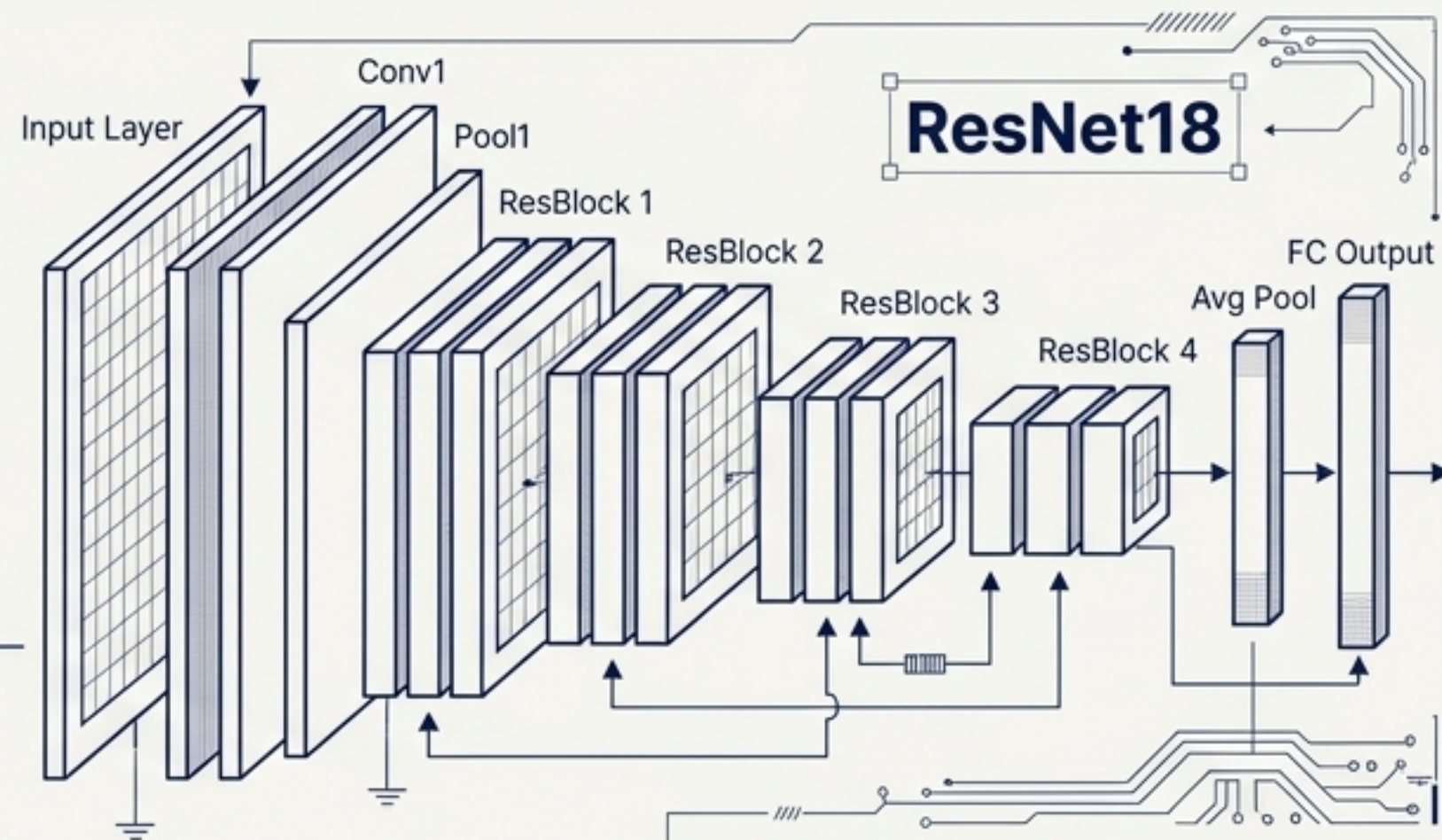
Number of Images by Gender and Style (training)



Highest volume training classes: Male Normcore & Female Sportivecasual.

Mission 1: Visual Feature Extraction

Transforming unstructured fashion photography into high-dimensional numerical vectors.



Key Objectives

1. Isolate the primary garment from complex backgrounds.

2. Train a ResNet18 convolutional neural network for accurate style classification.

3. Extract the final layer embeddings for downstream similarity calculations.

Hypothesis Testing: Preprocessing Optimization

YOLO object detection provided superior noise reduction over pixel-level segmentation, establishing a cleaner dataset for the ResNet18 pipeline.

Original Input



Image Segmentation



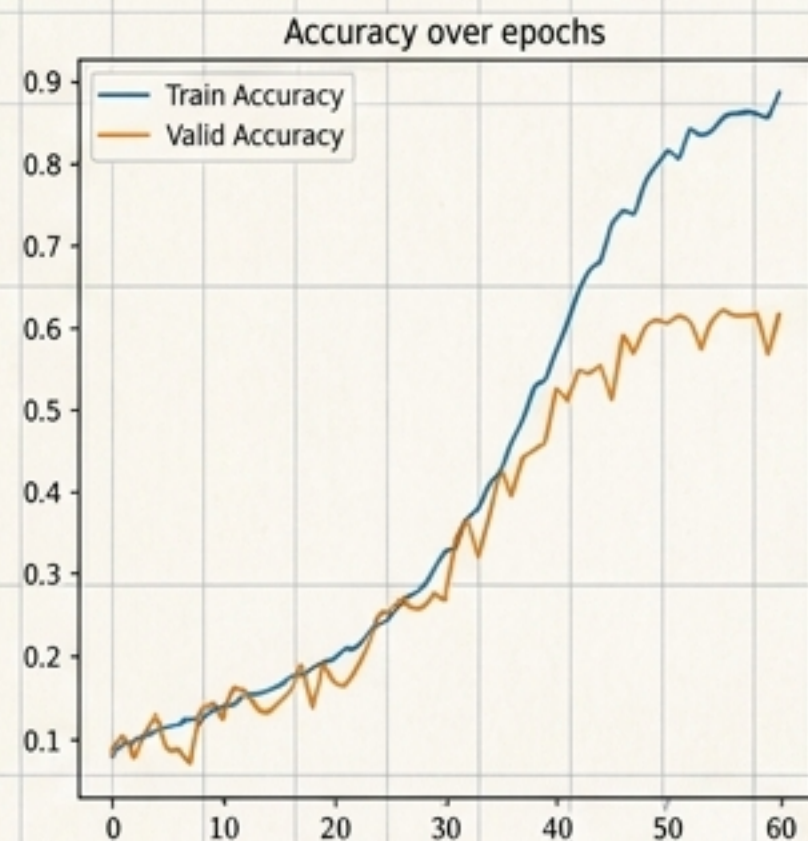
Method: Image Segmentation
Validation Accuracy: 51.64%
Verdict: Discarded (Too sensitive to background noise)

YOLO Object Detection



Method: YOLO Object Detection
Validation Accuracy: 61.51%
Verdict: Selected (Stable, precise garment isolation)

ResNet18 Experimental Parameters



Loss Function

CrossEntropyLoss() :
Optimal for multi-class
classification.

Optimizer

Adam (LR: 0.001) :
Balances speed and
stable convergence.

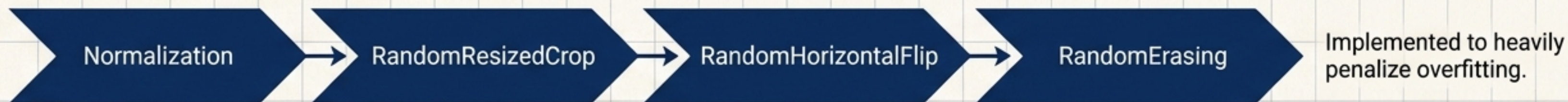
Batch Size

32 : Optimized for
memory efficiency.

Early Stopping

Patience = 20 : Extended
from 10 to capture late-
stage convergence over
200 max epochs.

Data Augmentation Module



Explainable AI (Grad-CAM)

Visualizing the network's attention confirms the 61.51% accuracy stems from genuine garment features, not spurious background correlations.

Male Subject: Complex Pattern Focus



Algorithm focuses intensely on complex geometric fabric patterns.

Female Subject: Silhouette & Boundary Detection



Model ignores the visually loud background wall and actively traces the garment's outer structural silhouette.

MISSION 2: UNDERSTANDING HUMAN PREFERENCE

By intersecting raw JSON survey arrays with validated image IDs, we synthesize top-tier respondent profiles for behavioral modeling.



	RESPONDENT ID	TRAINING PREFERRED	TRAINING NON-PREFERRED	VALIDATION PREFERRED	VALIDATION NON-PREFERRED
1	64747	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
2	65139	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
3	63405	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
4	64561	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
5	64348	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
...	...	...	...	...	...
100	64598	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
137	28571	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
108	35514	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
109	63316	W_04972_90_Sitsch_Wjpg, W_28783_10_sportiv...	W_02498_50_feminine_Wjpg, W_37168_70_hip...	W_26998_70_military_Wjpg, W_37481_70_military...	W_02498_50_feminine_Wjpg, W_37168_70_hip...
100	64295	...	...	...	...

100 rows x 5 columns

Top 100 density user profiles successfully mapped to explicit visual behaviors.

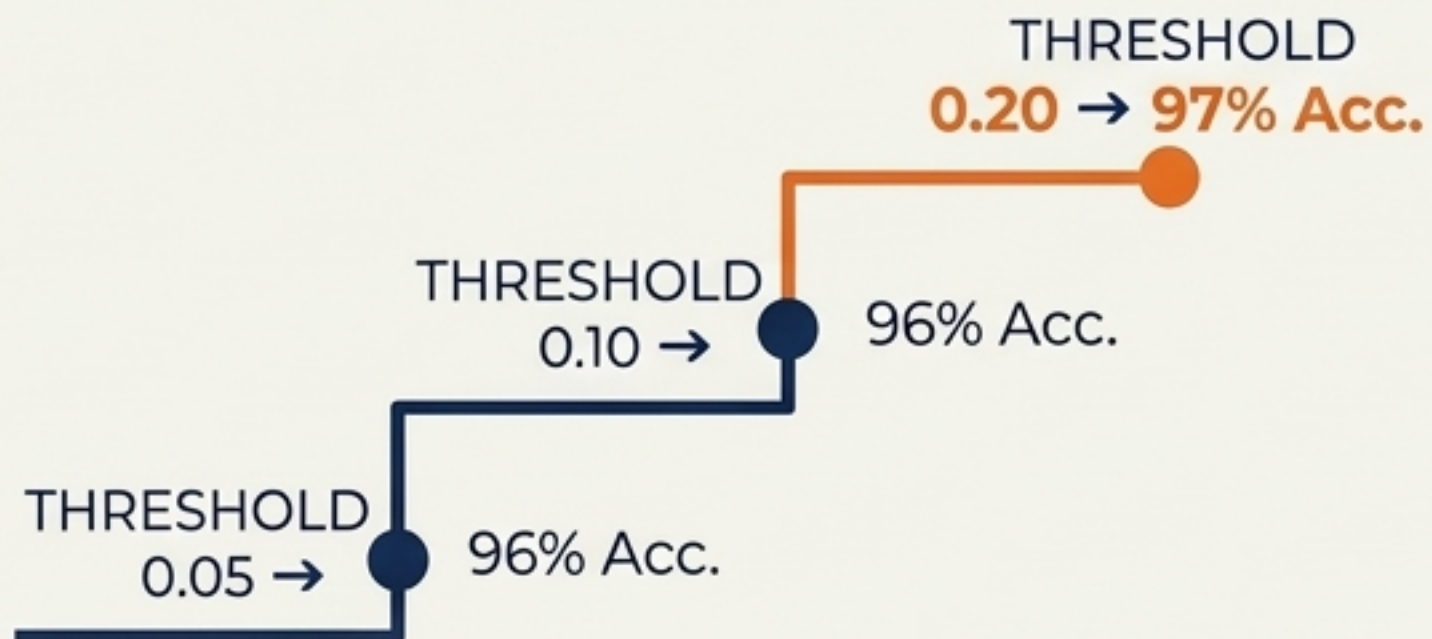
FEATURE SELECTION & BEHAVIORAL PREDICTION

Iterative tuning of **Cramer's V** isolated the most predictive categorical features, allowing the Random Forest to **predict user preference** with exceptional accuracy.

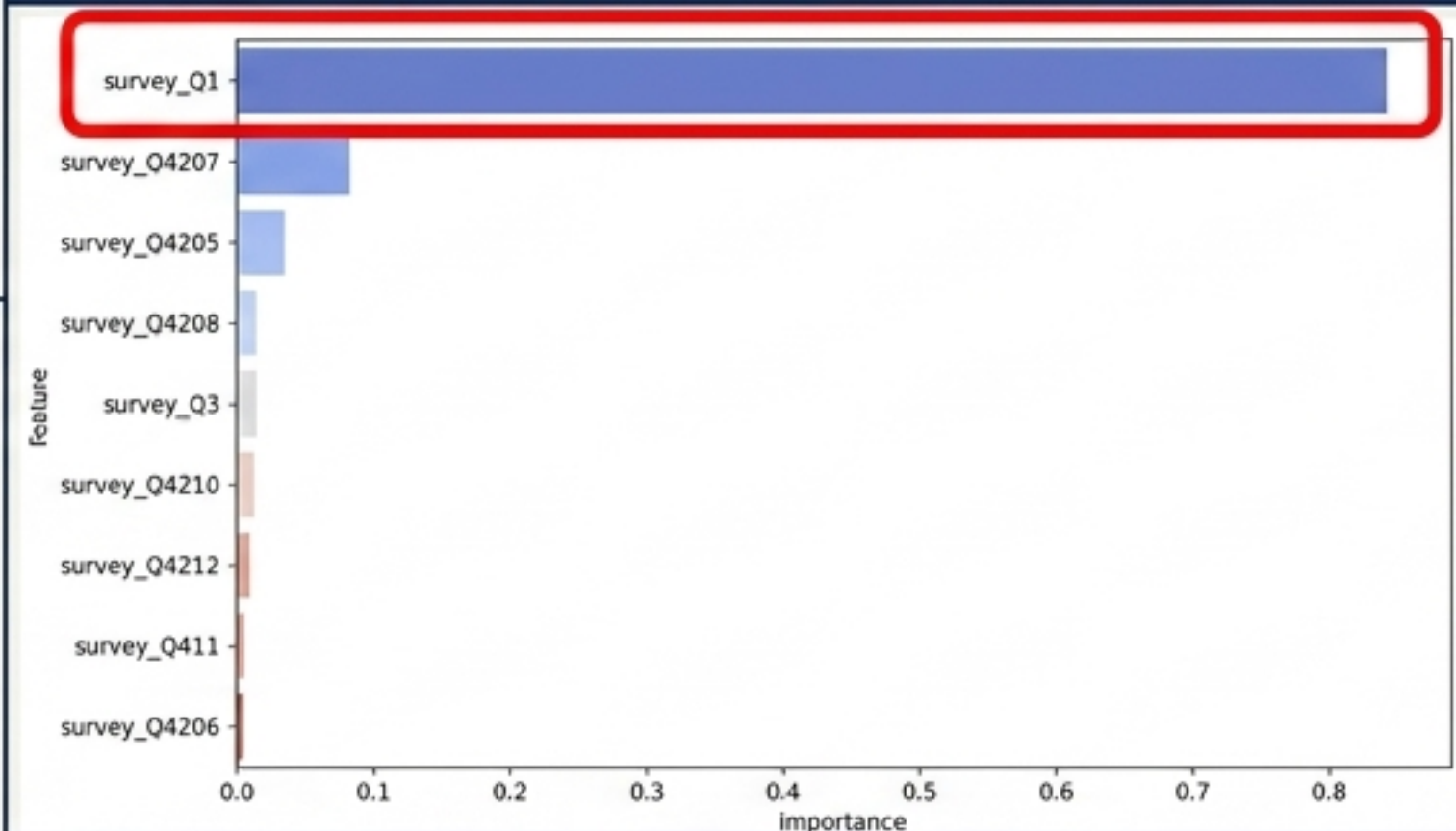
METHOD: CHI-SQUARE TEST

Evaluated categorical independence.

HYPERPARAMETER TUNING: CRAMER'S V THRESHOLD



FEATURE IMPORTANCES (RANDOM FOREST)



FINAL MODEL: RANDOM FOREST | 97% PREFERENCE PREDICTION ACCURACY

ARCHITECTURAL DECISION: RECOMMENDATION ENGINE

Selected **Item-Based** framework to ensure robust performance against data sparsity and to bypass the new-user cold-start problem.

ITEM-BASED CF (SELECTED FRAMEWORK)

Pros:

High stability, rapidly scalable, strictly solves the “cold-start” problem for new users.

Cons:

Limited cross-genre discovery.

VS

USER-BASED CF (DISCARDED FRAMEWORK)

Pros:

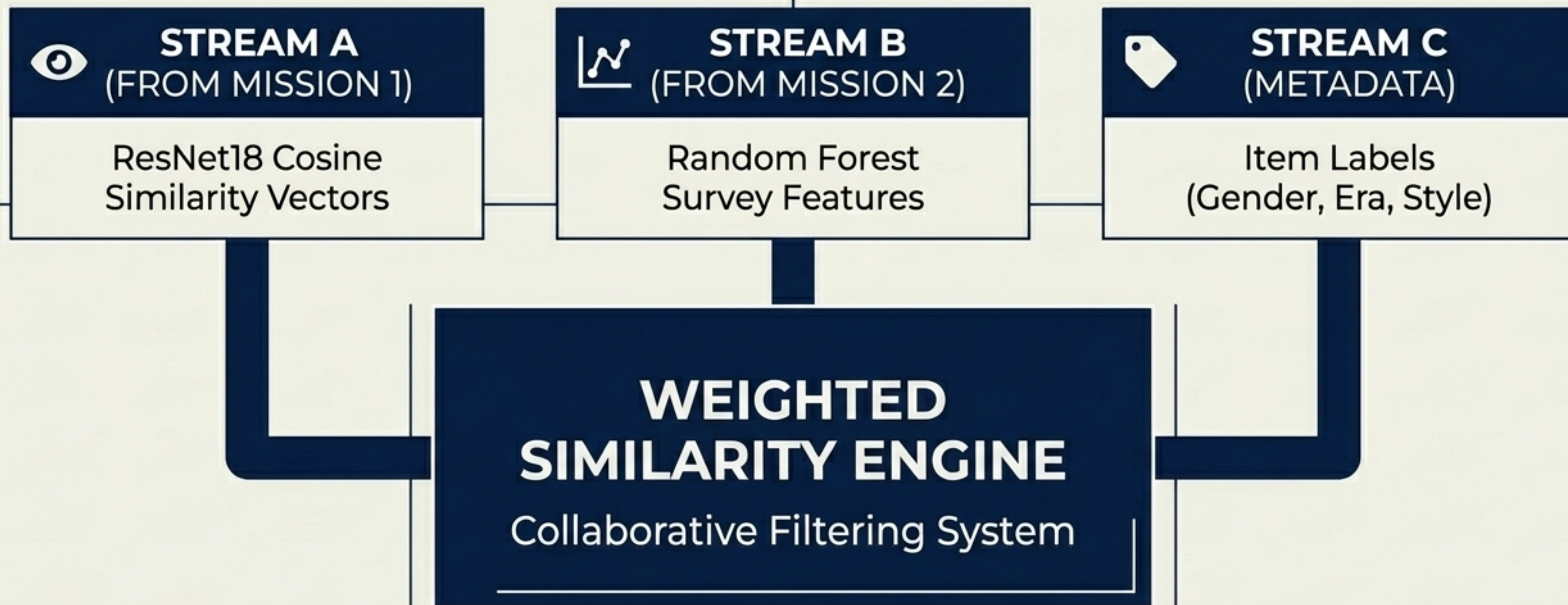
Deep personalization, easy interpretation via community trends.

Cons:

Severe scalability bottleneck as user base grows, highly susceptible to data sparsity.

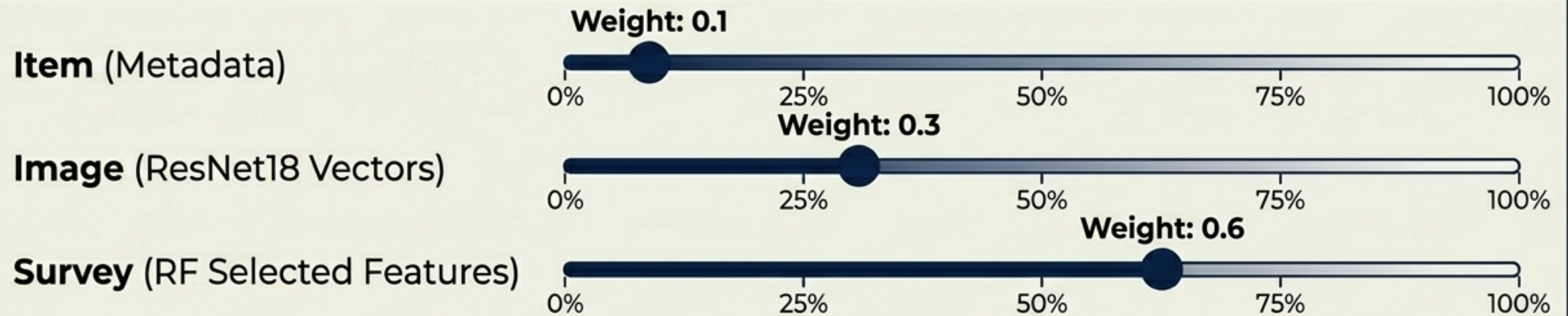
THE UNIFIED ARCHITECTURE

Disparate modalities—computer vision embeddings, algorithmic preference predictions, and categorical metadata—are mathematically fused into a single **Collaborative Filtering** engine.



THE WEIGHTING EQUATION

Empirical tuning revealed that user-declared survey features paired with deep visual embeddings yield the highest predictive accuracy.



TESTING ITERATIONS

0.2 Item / 0.5 Image / 0.3 Survey --> **Accuracy: 89.38%**

0.2 Item / 0.4 Image / 0.4 Survey --> **Accuracy: 90.01%**

0.1 Item / 0.5 Image / 0.4 Survey --> **Accuracy: 90.85%**

0.1 Item / 0.3 Image / 0.6 Survey --> Accuracy: 92.11% (Optimal Baseline)

A High-Fidelity, Multimodal Recommendation System.

Accuracy

92.11%

F1 Score

90.37%

Precision

88.89%

Recall

91.91%

By rigorously optimizing object detection, visual feature extraction, and behavioral survey modeling, the final ensemble framework successfully bridges the gap between **raw pixel** data and highly accurate, **personalized human** fashion preference.